

Matplotlib & Seaborn Practice Questions - E-commerce Sales Dataset

Dataset Overview

You'll be working with an e-commerce sales dataset containing 2000 transactions with the following columns:

- `transaction_id`: Unique transaction identifier
 - `date`: Transaction date
 - `category`: Product category (Electronics, Clothing, Books, etc.)
 - `price`: Unit price of the product
 - `quantity`: Number of items purchased
 - `subtotal`: Price × Quantity
 - `discount_percent`: Discount percentage applied
 - `discount_amount`: Dollar amount of discount
 - `total_amount`: Final amount after discount
 - `customer_segment`: Premium, Regular, or Budget
 - `region`: Geographic region (North, South, East, West, Central)
 - `sales_channel`: Online, Retail Store, or Mobile App
 - `customer_satisfaction`: Rating from 1-5
 - `return_customer`: Boolean indicating if customer has purchased before
 - `month`: Month number (1-12)
 - `quarter`: Quarter (Q1-Q4)
 - `day_of_week`: Day name (Monday-Sunday)
 - `is_weekend`: Boolean indicating weekend transaction
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BASIC MATPLOTLIB QUESTIONS (Questions 1-15)

1. Basic Line Plot

Create a line plot showing total sales amount by month. Add proper title, axis labels, and grid.

2. Bar Chart

Create a bar chart showing total revenue by product category. Rotate x-axis labels if needed.

3. Histogram

Create a histogram of customer satisfaction scores with 10 bins. Add appropriate title and labels.

4. Scatter Plot

Create a scatter plot of price vs. customer satisfaction. Add a trend line using `numpy.polyfit`.

5. Multiple Subplots

Create a 2x2 subplot layout showing:

- Top left: Sales by region (bar chart)
- Top right: Quantity distribution (histogram)
- Bottom left: Price vs total_amount (scatter)
- Bottom right: Sales by day of week (bar chart)

6. Pie Chart

Create a pie chart showing the distribution of sales channels. Include percentages and explode the largest segment.

7. Box Plot

Create a box plot comparing total_amount across different customer segments.

8. Stacked Bar Chart

Create a stacked bar chart showing sales by region, with each bar divided by customer segment.

9. Time Series Plot

Create a time series plot showing daily total sales over time. Use date on x-axis.

10. Customized Styling

Recreate question 1 but with custom colors, line styles, markers, and a professional theme.

11. Error Bars

Create a bar chart of average satisfaction by category with error bars showing standard deviation.

12. Log Scale

Create a scatter plot of price vs quantity using a log scale for the price axis.

13. Secondary Y-axis

Create a plot with sales volume (left y-axis) and average price (right y-axis) by month.

14. Annotation

Create a scatter plot of price vs satisfaction and annotate the top 5 highest-priced transactions.

15. Custom Legend

Create a scatter plot colored by customer segment with a custom legend positioned outside the plot area.

INTERMEDIATE MATPLOTLIB QUESTIONS (Questions 16-25)

16. 3D Scatter Plot

Create a 3D scatter plot with price, quantity, and satisfaction as the three axes.

17. Polar Plot

Create a polar plot showing sales by month (use month as angle, sales as radius).

18. Filled Areas

Create an area plot showing cumulative sales over time for each sales channel.

19. Custom Color Maps

Create a scatter plot of price vs satisfaction with points colored by total_amount using a custom colormap.

20. Multiple Y-axes

Create a plot with three different metrics on separate y-axes: total sales, average satisfaction, and transaction count by month.

21. Broken Axis

Create a bar chart of sales by category with a broken y-axis to handle outliers.

22. Dashboard Layout

Create a comprehensive dashboard with 6 different visualizations using GridSpec for custom layout.

23. Interactive Elements

Create a plot with clickable legend items that show/hide data series.

24. Custom Markers

Create a scatter plot using different marker shapes for each customer segment and sizes based on quantity.

25. Animation

Create an animated bar chart showing how sales by category change over quarters.

BASIC SEABORN QUESTIONS (Questions 26-40)

26. Basic Distributions

Use `sns.histplot()` to show the distribution of `total_amount` with KDE overlay.

27. Categorical Plots

Create a box plot using `sns.boxplot()` showing satisfaction scores across different sales channels.

28. Correlation Heatmap

Create a correlation heatmap of all numerical variables using `sns.heatmap()`.

29. Pair Plot

Create a pair plot of `price`, `quantity`, `satisfaction`, and `total_amount`, colored by customer segment.

30. Count Plot

Use `sns.countplot()` to show the frequency of transactions by day of the week.

31. Strip Plot

Create a strip plot showing price distribution across different categories with jitter.

32. Violin Plot

Create violin plots comparing `total_amount` distributions across regions.

33. Point Plot

Use `sns.pointplot()` to show average satisfaction by category with confidence intervals.

34. Joint Plot

Create a joint plot of `price` vs `total_amount` with marginal distributions.

35. Facet Grid

Use `sns.FacetGrid()` to create separate scatter plots of `price` vs `satisfaction` for each customer segment.

36. Regression Plot

Create a regression plot of `price` vs `total_amount` using `sns.regplot()`.

37. Categorical Heatmap

Create a heatmap showing average satisfaction by category and region.

38. Swarm Plot

Use `sns.swarmplot()` to show satisfaction distribution by sales channel.

39. Bar Plot with Error Bars

Create a bar plot showing average total_amount by category with confidence intervals.

40. Distribution Comparison

Use `sns.distplot()` or `sns.histplot()` to compare satisfaction distributions between return and new customers.

ADVANCED SEABORN QUESTIONS (Questions 41-60)

41. Multi-level Grouping

Create a grouped bar plot showing average sales by region and customer segment.

42. Custom Color Palettes

Recreate question 26 using a custom color palette that matches your company branding.

43. Statistical Annotations

Create box plots of total_amount by category and add statistical significance annotations.

44. Complex Heatmaps

Create a heatmap showing sales performance (rows: categories, columns: months) with custom annotations.

45. Ridge Plot

Create ridge plots (using multiple `sns.kdeplot()`) showing price distributions for each category.

46. Cluster Map

Use `sns.clustermap()` to create a clustered heatmap of correlations between numerical variables.

47. Time Series with Seaborn

Create a time series plot using seaborn showing sales trends with confidence bands.

48. Multi-panel Figure

Use `sns.FacetGrid()` to create a 2x3 grid showing satisfaction distributions for each sales channel and customer segment combination.

49. Custom Annotations

Create a correlation heatmap with custom annotations showing both correlation values and significance levels.

50. Advanced Pair Plot

Create a pair plot with custom diagonal plots (KDE instead of histograms) and off-diagonal regression lines.

51. Seasonal Decomposition Visualization

Decompose the time series into trend, seasonal, and residual components and visualize them.

52. Cohort Analysis Heatmap

Create a cohort analysis heatmap showing customer retention rates over time.

53. Funnel Analysis

Create a funnel chart showing conversion rates across different stages of the customer journey.

54. Geographic Heatmap

Create a simulated geographic heatmap showing sales intensity by region.

55. Customer Segmentation Visualization

Use multiple plots to visualize customer segments based on RFM analysis (Recency, Frequency, Monetary).

INTEGRATION & STYLING QUESTIONS (Questions 56-70)

56. Matplotlib + Seaborn Hybrid

Combine matplotlib's flexibility with seaborn's statistical plotting in a single complex figure.

57. Custom Themes

Create and apply a custom seaborn theme that matches your organization's style guide.

58. Publication-Ready Plots

Create publication-quality plots with proper sizing, fonts, and formatting for academic papers.

59. Interactive Dashboard

Combine multiple visualizations into an interactive dashboard using matplotlib widgets.

60. Subplot Mosaic

Use matplotlib's subplot_mosaic to create a complex layout with different sized subplots.

61. Color Accessibility

Recreate any previous plot ensuring it's accessible to colorblind users.

62. Multi-language Support

Create plots with axis labels and titles in multiple languages.

63. Dynamic Legends

Create legends that automatically adjust based on the data being displayed.

64. Custom Error Visualization

Create custom error visualizations for business metrics with confidence intervals.

65. Comparative Analysis

Create side-by-side comparisons of the same data using different visualization techniques.

66. Performance Metrics Dashboard

Create a comprehensive business dashboard showing KPIs with appropriate chart types.

67. Storytelling with Data

Create a sequence of plots that tell a coherent story about customer behavior.

68. A/B Testing Visualization

Visualize A/B testing results with appropriate statistical indicators.

69. Anomaly Detection Visualization

Create plots that highlight anomalies or outliers in the sales data.

70. Comprehensive Report

Create a multi-page visual report combining all the techniques learned, suitable for executive presentation.

BONUS CHALLENGES

Challenge A: Real-time Dashboard

Create a dashboard that updates in real-time as new data points are added.

Challenge B: Machine Learning Integration

Visualize the results of clustering algorithms applied to customer segments.

Challenge C: Web Integration

Export your visualizations in formats suitable for web integration (SVG, interactive HTML).

Challenge D: Mobile Optimization

Create responsive visualizations that work well on mobile devices.

Challenge E: Automated Reporting

Create a system that automatically generates weekly/monthly visual reports.

Getting Started Code Template

```
python
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from datetime import datetime
import warnings
warnings.filterwarnings('ignore')

# Load the dataset
df = pd.read_csv('ecommerce_sales_data.csv')
df['date'] = pd.to_datetime(df['date'])

# Set style
plt.style.use('default') # or try 'seaborn-v0_8', 'ggplot', etc.
sns.set_palette("husl")

# Your solution code here
```

Tips for Success

1. **Start Simple:** Begin with basic plots and gradually add complexity
2. **Read Documentation:** Familiarize yourself with matplotlib and seaborn documentation
3. **Experiment:** Try different parameters and styling options
4. **Business Context:** Always consider what story your visualization tells
5. **Iterate:** Refine your plots based on feedback and best practices
6. **Save Your Work:** Keep a notebook with all your solutions for future reference

Evaluation Criteria

For each visualization, consider:

- **Clarity:** Is the message clear?
- **Accuracy:** Does it represent the data correctly?
- **Aesthetics:** Is it visually appealing?
- **Functionality:** Does it serve its intended purpose?
- **Code Quality:** Clean, readable, well-commented code

Good luck with your matplotlib and seaborn journey!